EA-PDF SOFTWARE ARCHITECTURE

Thinkbox.DIGITAL | Product and dependency overview | 13 November 2026

EA-PDF CREATOR

Java 21+ | Desktop and command line
2.2.0-beta.2

Email to preservation PDF
EML, MSG, PST, MBOX and Gmail. PDF 2.0 / PDF/A-4f, email metadata, document parts and attachments.

Output choices
Single messages, combined archives or collections. Optional accessibility features, search indexes and signing.

STANDALONE VALIDATOR

Java 21+ | Desktop and command line
2.2.0-beta.2

Checks
229 project rule definitions; PDF/A validation; selected ISO 32005 structure rules.

Scope
Embedded PDFs, attachments and source consistency. Reports PASS, FAIL or INCOMPLETE. No server required.

FULL AND LITE READERS

Chromium and Firefox | Local files
1.2.0-beta.2

Shared reading experience
PDF pages, message/folder navigation, attachments, text search, filtering and CSV/XMP export.

Lite
Same layout; no Conversations, Portfolio view, embedded indexes or large-archive optimizations. 32 MiB / 2,048 pages.

CREATOR LIBRARIES

PDF creation and metadata
PDFBox / XMPBox 3.0.8. OpenHTMLtoPDF 1.1.113-eapdf.1 with four maintained patches.

Email and content
Jakarta Mail / Angus Mail, java-libpst, Outlook Message Parser, POI 6.0.0, jsoup 1.24.1, RTF-to-HTML and jutf7.

Other integrations
Bouncy Castle 1.87; Google Gmail / OAuth clients; Gson 2.15.0; metadata-extractor 2.22.0.

VALIDATION LIBRARIES

veraPDF engines and project checks
Core / XMP 1.31.56; parser 1.31.71; PDF model 1.31.14; validation model 1.31.221. Local storage/resource patches remain.

PDF and email inspection
PDFBox 3.0.8; POI 6.0.0; Gson 2.15.0; Jakarta Mail / Angus Mail; project EA-PDF checks.

Optional development tool
Python CLI for regression tests and diagnostics. The web workbench and its deployment code are removed.

READER FOUNDATION

PDF engine and shared hooks
PDF.js 6.5.438. Generic hooks expose document parts / DPM, associated files, metadata and collection membership.

Application layer
TypeScript 5.9.3; browser extensions. Full and Lite share maintained source; Lite is a reproducible standalone export.

Build tools only
Node 24.19.0; npm 11.17.0; Playwright 1.64.1. No Java or Python needed by the extensions.

High-level overview: transitive libraries and some helpers are omitted. Rule counts are project definitions, not a claim of complete specification coverage.

Preserve original mailboxes separately. Validation has defined limits; PDF/UA machine checks do not replace human accessibility review.

Source and maintenance notes: docs/ARCHITECTURE.md | github.com/thinkbox-digital/eapdf